Sets & Relation: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Sets, representation and types

KEY POINTS

- Set = well-defined collection of distinct objects; $n(A)$ = number of elements.
- Roster $\{1, 2, 3\}$; set-builder $\{x : P(x)\}$.
- Finite / infinite; singleton $n = 1$; null ϕ, $n = 0$.
- Equal: same elements. Equivalent: same number of elements.

STEPS

1. Set given by a condition: solve it, keep only values in the universe $(\mathbb{N}, \mathbb{Z}, \mathbb{R})$, then list or count.

Watch out

$\{\phi\}$ and $\{0\}$ are not empty; $\phi \neq \{\phi\}$.

2. Subsets and power set

Formulas

$$\#\text{subsets} = 2^n, \quad \#\text{proper} = 2^n - 1, \quad n(P(A)) = 2^n$$

KEY POINTS

- $A \subseteq B$: every element of A is in B; then $B \supseteq A$ (superset).
- $\phi \subseteq A$ and $A \subseteq A$ always.
- Containing k fixed elements: 2^{n-k}; disjoint ordered pairs (Y, Z): 3^n.

Watch out

$a \in A$ but $\{a\} \subseteq A$: do not mix $\in$ and $\subseteq$.

3. Operations and laws

Formulas

$$A - B = A \cap B', \qquad A \,\Delta\, B = (A - B) \cup (B - A)$$
$$(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

KEY POINTS

- Idempotent, identity $(A \cup \phi = A,\ A \cap U = A)$, commutative, associative laws.
- Disjoint: $A \cap B = \phi$. $A \cup B = A \cup C$ and $A \cap B = A \cap C \Rightarrow B = C$.
- Venn diagram: 3 sets make 8 regions; read the statement region by region.

Watch out

Interval sets: check the end points when taking $A - B$, $B - A$.

4. Number of elements

Formulas

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$
$$\max(0, n(A) + n(B) - n(U)) \leq n(A \cap B) \leq \min(n(A), n(B))$$

KEY POINTS

- Only $A = n(A) - n(A \cap B)$; none $= n(U) - n(A \cup B)$.
- Exactly two $= \sum n(A \cap B) - 3n(A \cap B \cap C)$.
- Multiples of k in $1..N$: $[N/k]$; "a and b" means multiples of $\text{lcm}(a, b)$.

STEPS

1. Draw the Venn diagram and fill the centre first ($A \cap B \cap C$), then the "two only" regions, then "one only".

Watch out

"Only A" is not $n(A)$; percentages: take $n(U) = 100$.

5. Cartesian product

Formulas

$$A \times B = \{(a, b) : a \in A,\ b \in B\}, \qquad n(A \times B) = mn$$
$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

KEY POINTS

- $\times$ distributes over $\cup$, $\cap$, $-$.
- Common elements of $A \times B$ and $B \times A$: $n(A \cap B)^2$.
- Subsets of $A \times B$ with exactly r elements: ${}^{mn}C_r$.

Watch out

$A \times B \neq B \times A$ in general; $(a, b) \neq (b, a)$.

6. Relation: domain, range, counting

Formulas

$$R \subseteq A \times B, \qquad \#\{R : A \to B\} = 2^{mn}, \qquad \#\{R \text{ on } A\} = 2^{n^2}$$

KEY POINTS

- Domain = first entries; range = second entries; co-domain = whole B.
- $(a, b) \in R$: b is an image of a, a a pre-image of b.
- Inverse $R^{-1} = \{(b, a)\}$: domain and range swap.

STEPS

1. Counting a relation given by a rule: fix the first entry, count the second, add.

7. Reflexive, symmetric, transitive

Formulas

$$\text{Ref} : (a, a) \in R\ \forall a; \quad \text{Sym} : aRb \Rightarrow bRa; \quad \text{Trans} : aRb,\ bRc \Rightarrow aRc$$
$$\text{Anti-symmetric} : aRb,\ bRa \Rightarrow a = b$$

STEPS

1. Reflexive: put $y = x$ in the rule.
2. Symmetric: swap x and y.
3. Transitive: prove, or find a counter-example with 0, ± 1, $\frac{1}{2}$, irrational numbers.
4. Equivalence = all three; equivalence classes form a partition.

KEY POINTS

- Void relation: symmetric and transitive; universal and identity: equivalence.
- Factorise rules like $x^2 - 4xy + 3y^2 = 0 \Rightarrow x = y$ or $x = 3y$ before testing.

Watch out

The set matters: "divides" is anti-symmetric on $\mathbb{N}$ but not on $\mathbb{Z}$; 0 breaks reflexivity on $\mathbb{Z}$.

8. Counting relations and closures

Formulas

$$\text{Ref} : 2^{n(n-1)}, \quad \text{Sym} : 2^{\frac{n(n+1)}{2}}, \quad \text{Ref and Sym} : 2^{\frac{n(n-1)}{2}}, \quad \text{Equiv} : B_n$$

KEY POINTS

- Bell numbers: $B_1 = 1,\ B_2 = 2,\ B_3 = 5,\ B_4 = 15$.
- Smallest equivalence containing R: blocks of connected elements, $\sum s_i^2$ pairs (s_i = block sizes).

STEPS

1. Make reflexive: add missing (a, a); symmetric: add missing (b, a); transitive: add (a, c) for each chain, repeat.
2. Answer = pairs in the new relation − pairs already in R.

Watch out

After adding mirrors for symmetry, re-check transitivity (it may force (a, a)).

9. Common mistakes checklist

STEPS

1. Universe of the set ($\mathbb{N}$ starts at 1; $\mathbb{Z}$ includes negatives and 0).
2. Count each region of a Venn diagram once.
3. Non-empty subsets $= 2^n - 1$; "at least 3 elements" = total − sizes 0, 1, 2.
4. Reflexive needs every (a, a), not just some.
5. Relation on $A \times B$: its elements are pairs of pairs.